

Total No. of Questions : 8]

PC2494

SEAT No. :

[Total No. of Pages : 5

[6354]-623

**B.E. (MECHANICAL)
OPERATIONS RESEARCH**

(2019 Pattern) (Semester - VII) (Elective - IV) (402045 D)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Answers in one answer books.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Use Graphical method to solve the following LPP.

[10]

$$\text{Maximize } Z = 1170X_1 + 1110X_2$$

Subject to Constraints

$$9X_1 + 5X_2 \geq 500$$

$$7X_1 + 9X_2 \geq 300$$

$$5X_1 + 3X_2 \leq 1500$$

$$7X_1 + 9X_2 \leq 1900$$

$$2X_1 + 4X_2 \leq 1000$$

$$X_1, X_2 \geq 0$$

b) Explain in brief

[7]

- i) Objective function
- ii) Constraint surface
- iii) Feasible and infeasible points
- iv) Optimum solution

OR

P.T.O.

Q2) a) Use Simplex method to solve the following LPP

$$\text{Maximize } Z = 80 X_1 + 55 X_2,$$

Subject to Constraints

$$4X_1 + 2X_2 \leq 40$$

$$2X_1 + 4X_2 \leq 32$$

$$X_1, X_2 \geq 0$$

[10]

b) Explain in brief

i) Limitations of graphical method

[3]

ii) Conversion of Primal to Dual in LPP problem

[4]

Q3) a) Discuss in brief the methodology used in Hungarian method to solve the any assignment problems for getting optimum solution. [8]

b) Find out the initial feasible solution by North West corner method. [10]

From	To	supply
2	11 10 3 7	4
1	4 7 2 1	8
3	1 4 8 12	9
Demand	3 3 4 5 6	

OR

Q4) a) An airline company has drawn up a new flight schedule that involves five flights. To assist in allocating five pilot to the five flights, it has asked them to state their preference scores by giving each flight a number out of 10. The higher the number, the greater is the preference. A few of these flights are unsuitable to some pilots, owing to domestic reasons. These have been marked with "X" [12]

		Flight Number				
		I	II	III	IV	V
Pilot	A	8	2	X	5	4
	B	10	9	2	8	4
	C	5	4	9	6	X
	D	3	6	2	8	7
	E	5	6	10	4	3

What should be the allocation of the pilots in order to meet maximum preference?

b) Differentiate between assignment and transportation problem.

[6]

Q5) a) A small project involves 7 activities and their times estimates are listed in the following table. Activities are identified by their beginning [12]

- i) and ending
- ii) node numbers.

Activities (i-j)	Estimated Duration (Weeks)		
	Optimistic	Most likely	Pessimistic
1-2	1	1	7
1-3	1	4	7
1-4	2	2	8
2-5	1	1	1
3-5	2	5	14
4-6	2	5	8
5-6	3	6	15

- i) Draw the network diagram of the activities in the projects.
- ii) Find expected duration and variance for each activity. What is the expected project length?
- iii) Calculate the variance and standard deviation of the project length. What is the probability that the project will be completed:
 - 1) At least 4 weeks earlier than expected time.
 - 2) No more than 4 weeks later than expected time.

Given

Z (0-Z)	1.33
Probability	0.4082

b) Explain looping and Dangling errors in the network. [6]

OR

- Q6) a)** A taxi owner estimates from his past records that the costs per year for operating a taxi whose purchase price when new is Rs. 60,000 are as given below :

Year	1	2	3	4	5
Operating Cost (Rs.)	10,000	12,000	15,000	18,000	20,000

After 5 years, the operating cost is Rs. 6000*K where K = 6,7,8,9 and 10 (K denotes the age of in years) If the resale value decreases by 10% of purchase price each year, what is the best replacement policy? Cost on money is zero. **[12]**

- b) Differentiate between CPM and PERT. **[6]**

- Q7) a)** A dentist scheduled all his patients for 30 minute appointments. Some of the patients take more or less than 30 minutes depending on the type of dental work to be done. The following summary shows the various categories of work, their probability and time actually needed to complete the work:

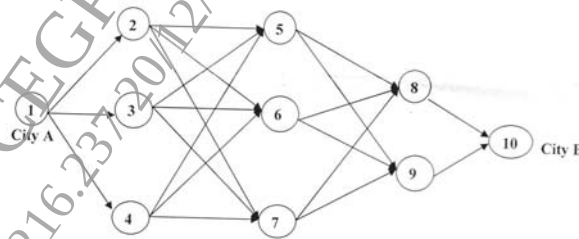
Category of service	Time required in Minute	Probability
Filling	45	0.40
Crown	60	0.15
Cleaning	15	0.15
Extraction	45	0.10
Check up	15	0.20

Simulate the dentist's clinic for four hours and determine the average waiting time for the patients as well as the idleness of the doctor. Assume that all the patients show up at the clinic at exactly their scheduled arrival time starting at 8:00 a.m. Use the following sequence of random numbers to simulate the above problem. Random Numbers: 40. 82. 11,34,25,66, 17,79. **[12]**

- b) Write short note on Monto Carlo Simulation. **[5]**

OR

- Q8) a)** A salesman located in a city A decided to travel to city B. He knew the distances of alternative routes from city A to city B. He then drew a highway network map as shown in following figure. The city of origin A, is city 1. The destination city B is city 10. Other cities through which the salesman will have to pass through are numbered 2 to 9. The arrow representing routes between cities and distances in kilometres are located on each route. The salesman problem is to find the shortest route that covers all the selected cities from A to B. The time for each activity is given in the table. (Solve by using Dynamic programming). **[12]**



Activity	Duration	Activity	Duration
1-2	4	4-5	6
1-3	6	4-6	10
1-4	3	4-7	5
2-5	7	5-8	4
2-6	10	5-9	8
2-7	5	6-8	3
3-5	3	6-9	7
3-6	8	7-8	8
3-7	4	7-9	4
		8-10	7
		9-10	9

- b)** Explain in brief various steps involved in the simulation. **[5]**

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